Intraday Delta Hedging: Backtest Results and Possible Improvements

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1 Setup and Benchmarks

All results below come from the backtesting framework of notebooks 04 (static bands) and 05 (dynamic bands): 102 trading days (January 2 – May 29, 2026), 32,127 simulated ES option executions, book delta and gamma repriced with Black-76 at every 5-minute bar, TWAP/VWAP execution, the spread-plus-impact execution-cost proxy, and delta risk measured as the RMS of realized residual-delta P&L per bar. The cost model incorporates two refinements from the `Timmy/strategy-comparison` branch: the impact level $\alpha = 0.055$ is anchored at ≈ 0.5bp of notional impact for a 1-lot at median ES ADV, and any fill on the final bar – in particular the forced end-of-day flatten – is costed as a Market-On-Close auction execution (zero spread, impact scaled by 0.15). Configurations are scored by the standardized loss

$$L_{\mathrm{std}} = \widetilde{C}_{\mathrm{exec}} + \lambda\, \widetilde{C}_{\mathrm{risk}}, \qquad \lambda = 1,$$

normalizing cost by Full Hedge and risk by Zero Hedge.

Benchmark	**Exec. cost**	**Risk (RMS/bar)**	L_{std}
Zero Hedge (flatten at close only)	\$235,240	\$24,801	1.058
Full Hedge (re-flatten every bar)	\$4,030,988	\$0	1.000

Table 1: Benchmark bounds over the 102-day backtest.

2 Static Band Results

The grid search over $\theta = (H, \epsilon, T_{\text{cooldown}}, e, \tau)$ selects

$$\theta^* = (H{=}25,\ \epsilon{=}10\%,\ T_{\text{cooldown}}{=}0,\ \text{TWAP},\ 60\,\text{min}) :$$

Strategy	**Exec. cost**	**Risk (RMS/bar)**	L_{std}
Static band θ^*	\$835,014	\$8,620	0.555

Table 2: Selected static-band strategy: −65% delta risk vs. Zero Hedge and −79% cost vs. Full Hedge (a saving of \$3.20M over the sample, ≈\$31,300 per day). Under the closing-auction refinement Zero Hedge's single cheap auction flatten makes it the cost floor by a wide margin, so the band's case rests on risk reduction and the combined objective, not on undercutting Zero Hedge's cost.

Robustness. On five re-seeded trade tapes (per-day bootstrap redrawing contract, side, and size), the selected strategy beats both benchmarks on the standardized objective on every seed. $H = 25$ with TWAP wins four of five seeds; one seed prefers a wide $H = 100$ band – consistent with the cheap auction close narrowing the gap between tight and wide bands. The finer parameters (ϵ, cooldown, horizon) reshuffle with the draw and should be treated as second-order.

3 Dynamic Band Results

The band is generalized to $H(t)$, with the static rule as the constant special case:

$$|\Delta(t)| > H(t)(1+\epsilon), \tag{1}$$

$$H(t) = H_0 \sqrt{\tfrac{R(t)}{R(0)}} \cdot \frac{1}{1 + c\,|\Gamma(t)|}, \tag{2}$$

where $R(t)$ is the number of bars remaining in the session and $\Gamma(t)$ the book gamma repriced each bar. The time-decay factor comes from the two-sided-flow argument (future arrivals can offset a delta of scale $\sqrt{R(t)}$ for free); the gamma factor from delta diffusing through the band at speed $d\Delta \approx \Gamma\,dS$ (proposal Eq. 32). Gamma enters undivided, so c is not dimensionless: it carries units of points per contract and the band halves at $|\Gamma| = 1/c$. This book's median $|\Gamma|$ is 0.0928 contracts/point, so the grid is $c \in \{5, 10, 20\}$.

The width trap. At matched $H_0 = 25$ the combined dynamic band improves L_{std} by $\approx 8\%$ over static – but dynamic bands are narrower *on average*, and under this cost proxy tighter is almost monotonically better: extending the static grid downward, an ultra-tight *static* band ($H_0 = 2$, $L_{\text{std}} = 0.495$) beats every dynamic configuration under full-flatten targeting. Naive comparisons at matched H_0 conflate band *timing* with band *tightness*.

Width-matched comparison. Setting a static band to each dynamic configuration's mean effective width (same horizon, method, ϵ, cooldown) isolates the timing information, and under the closing-auction cost model it *splits* the two mechanisms. Gamma conditioning keeps a genuine edge – roughly 2.5–4.0% of the standardized loss at moderate widths, positive on every re-seeded tape – because its rationale (delta diffuses through the band faster when gamma is high) does not depend on how the close is costed. The time-decay band's edge turns *negative*: its derivation leaned on the end-of-day flatten being expensive, and the auction discount removes exactly that penalty, so shrinking the band into the close pre-pays execution for nothing. The combined form inherits a diluted positive edge from its gamma component.

4 Band-Edge Targeting

Replacing the full flatten $Q_t = -\Delta(t)$ with the Whalley–Wilmott-style edge target

$$Q_t = -\operatorname{sign}(\Delta(t))\left(|\Delta(t)| - H(t)\right) \tag{3}$$

trades only the excess over the band and leaves the residual parked at the edge. Results (TWAP 60min, L_{std} change vs. full flatten):

H_0	5	10	25	50	100
Static band	+2.6%	+5.4%	+6.0%	+1.5%	−9.4%
Combined dynamic	+0.3%	+0.8%	+1.8%	+1.7%	+3.7%

Table 3: Improvement from edge targeting. Positive = better than full flatten.

Edge targeting is now the strongest single refinement: under the closing-auction cost model, parking residual delta at the band edge and carrying it to a cheap auction close is exactly the right trade, worth up to +6% of standardized loss at moderate static widths. It still backfires for wide static bands (too much risk parked at the edge under $\lambda = 1$), a failure mode the combined dynamic band removes – its shrinking width pulls the parked delta inward through the afternoon – keeping the stack uniformly positive across re-seeded tapes. The best overall configuration is **combined dynamic band + edge targeting** ($H_0 = 5$, $c = 20$: $L_{\text{std}} = 0.491$ vs. 0.495 for the best full-flatten configuration).

5 Cost-Model Discussion

The execution-cost proxy contains two components, both *size-proportional*: spread cost (linear in order size; half a tick per contract) and ADV-adjusted impact cost (superlinear, $\propto |q|^{1.5}$). Because both vanish as orders shrink, splitting a hedge into arbitrarily many small child orders is costless in the model, which is why ultra-tight bands (thousands of small triggers per sample) appear almost monotonically better. What the model *lacks* is a *fixed per-ticket cost* – a charge per order independent of size (exchange/clearing minimums, broker ticket fees, operational overhead). Any positive per-ticket cost penalizes high-frequency triggering, shifts the optimal band width to an interior value, and makes the band-design comparisons more decision-relevant. Re-optimizing the grid under a fixed charge per child order confirms this is a first-order effect on ES: at \$10 per order the ticket bill is already $\approx 9\%$ of execution cost, and by $\approx$\$100 per order the optimum leaves the tightest band (≈ 79 hedges/day) for a moderate one ($H = 25$, ≈ 24/day), moving further out as the charge rises. The very tight end of all results above should therefore be read as a model artifact, not a trading recommendation; the moderate-width regime is the operationally relevant one, and it is exactly where the dynamic-band and edge-targeting refinements earn their keep. (This is specific to ES's execution-cost scale. On the SPCX book a per-order charge is negligible against impact costs, and none is needed: that book's optimum is already an interior, wide band, so no per-ticket cost or order-splitting floor is required to avoid a boundary solution.)

6 Possible Improvements

1. **Fixed per-ticket cost.** Add C_{ticket} per child order to the cost proxy. Restores an interior optimal band width and is expected to increase the relative value of dynamic bands and edge targeting. Highest priority.

2. **Urgency-aware execution horizon.** Make $\tau(t) = \min(\tau_0, R(t))$ explicit (or decay τ with the same $\sqrt{R/R_0}$ factor as the band), so late-day hedges execute faster instead of being silently truncated by the close.

3. **Volatility-conditioned band.** The gamma factor captures how fast the book's delta moves per index point; a realized-volatility factor $1/(1+c_\sigma\,\hat{\sigma}(t)/\tilde{\sigma})$ would capture how many points the market is moving. Same plumbing as the gamma band.

4. **Derive the optimal band.** Solve a small dynamic program for $H^*(t)$ under a simplified arrival model calibrated to the tape, and benchmark the heuristic $\sqrt{R}$ and gamma forms against it.

5. **Historical VWAP weights.** Current VWAP uses same-day realized volumes (mild look-ahead); switching to a trailing 20-day time-of-day volume profile makes it implementable. TWAP's observed advantage would, if anything, grow.

6. **Calibrate α from TCA data.** The impact level is a sensitivity parameter, not a calibrated one; conclusions were checked across $\alpha \times \{0.5, \ldots, 4\}$, but real fill data would pin the dollar magnitudes.

7. **Out-of-sample validation.** Select parameters on the first $\sim$70 days and evaluate frozen on the remainder, quantifying the winner's-curse gap directly.

8. **Real flow structure.** The simulated tape draws sides i.i.d. 50/50, so arrivals contain no predictable structure beyond the $\sqrt{R}$ effect; real client flow is clustered and autocorrelated, which should increase the value of flow-conditioned (arrival-aware) bands and execution.